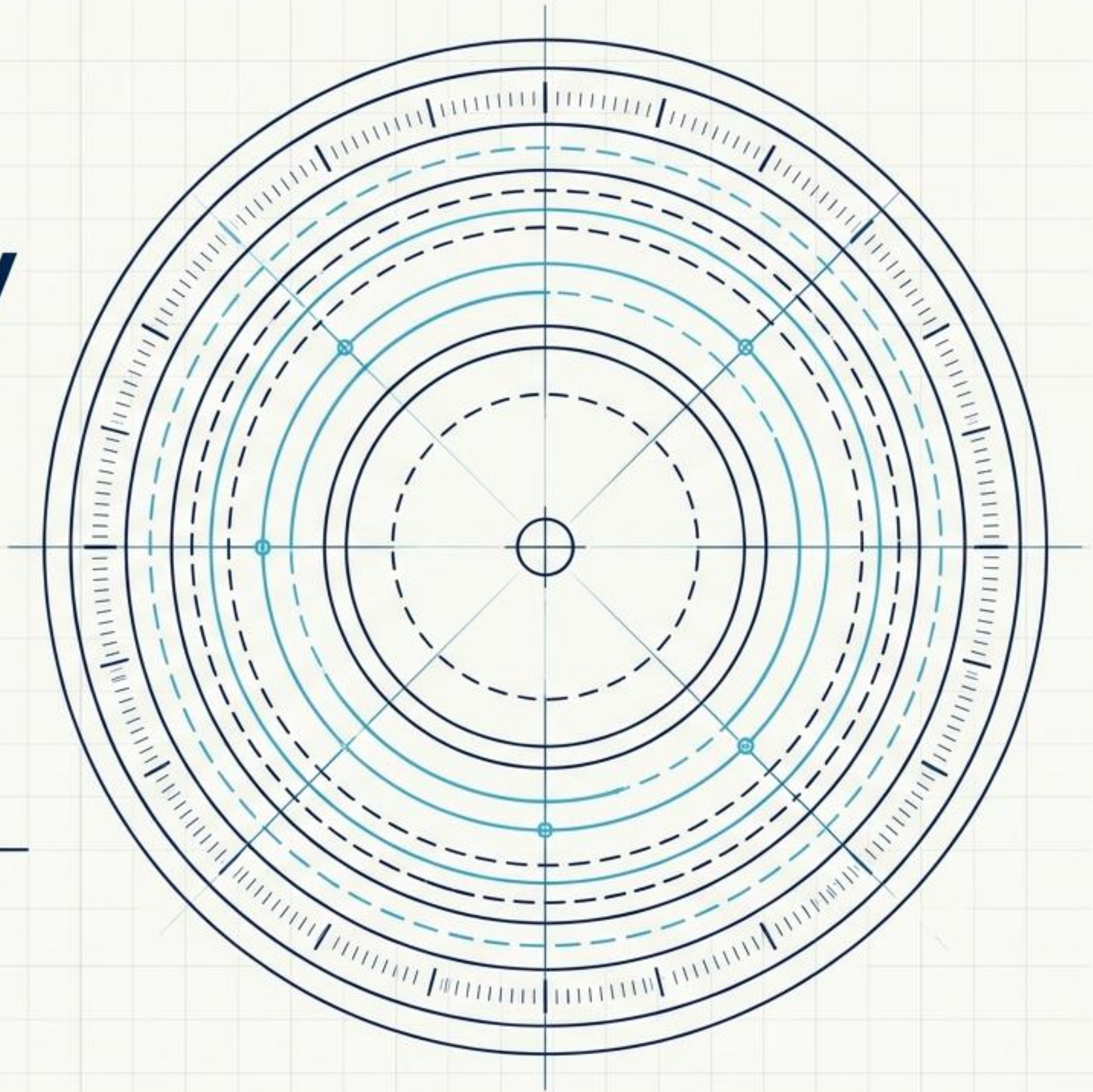


Certyfikat PMP: Project Efficiency Measurement

Mastering the
Monitoring &
Controlling Process
Ecosystem

dr Zbigniew Galar



The Project Navigator's Imperative

The core of the Monitoring and Controlling process group is continuous telemetry. It requires taking precise measurements and performing rigorous inspections to detect deviations from the project management plan.



1. Discover Variances

Identify gaps between planned and actual performance.

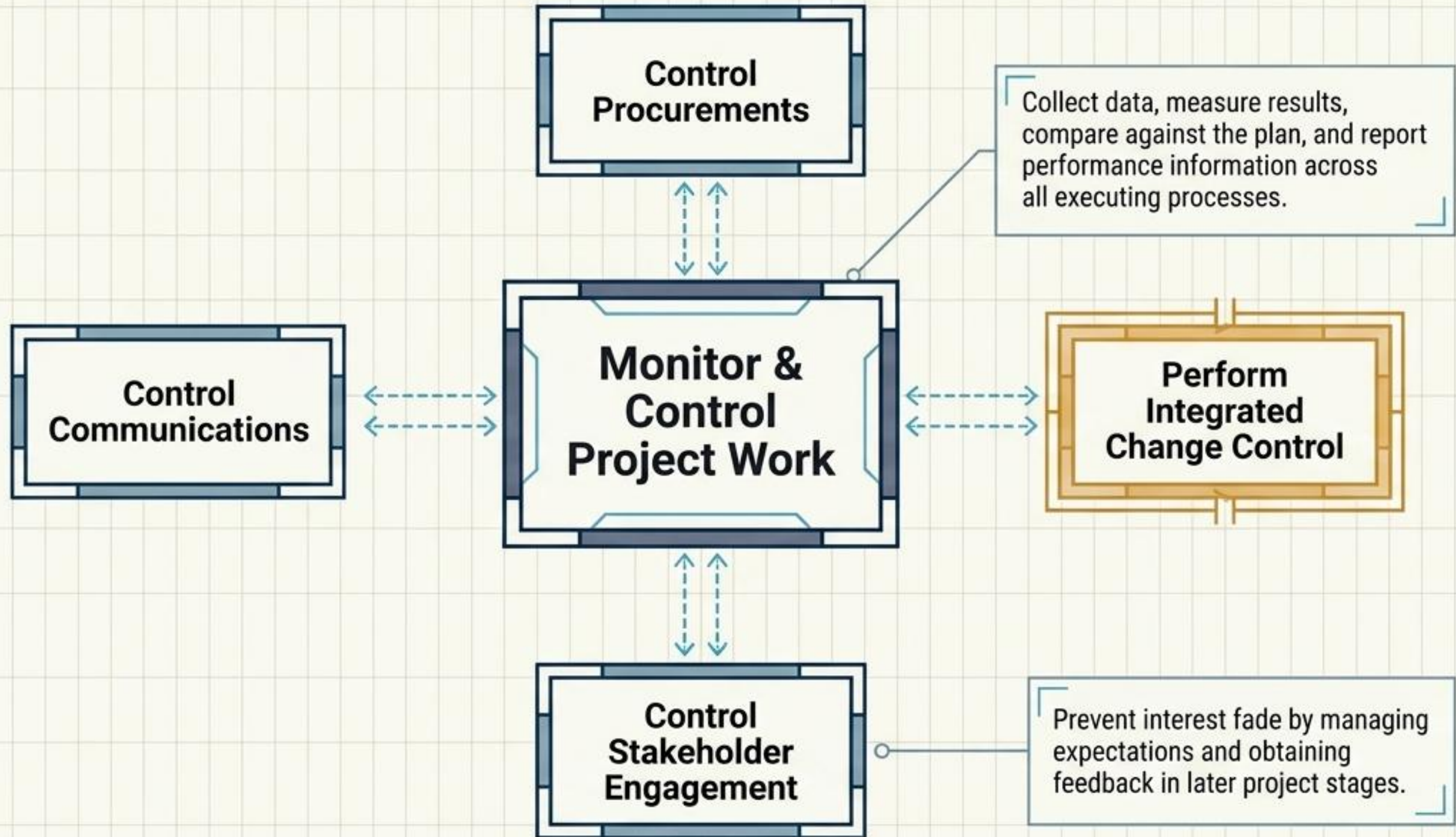
2. Take Corrective Action

Implement interventions to realign the project trajectory.

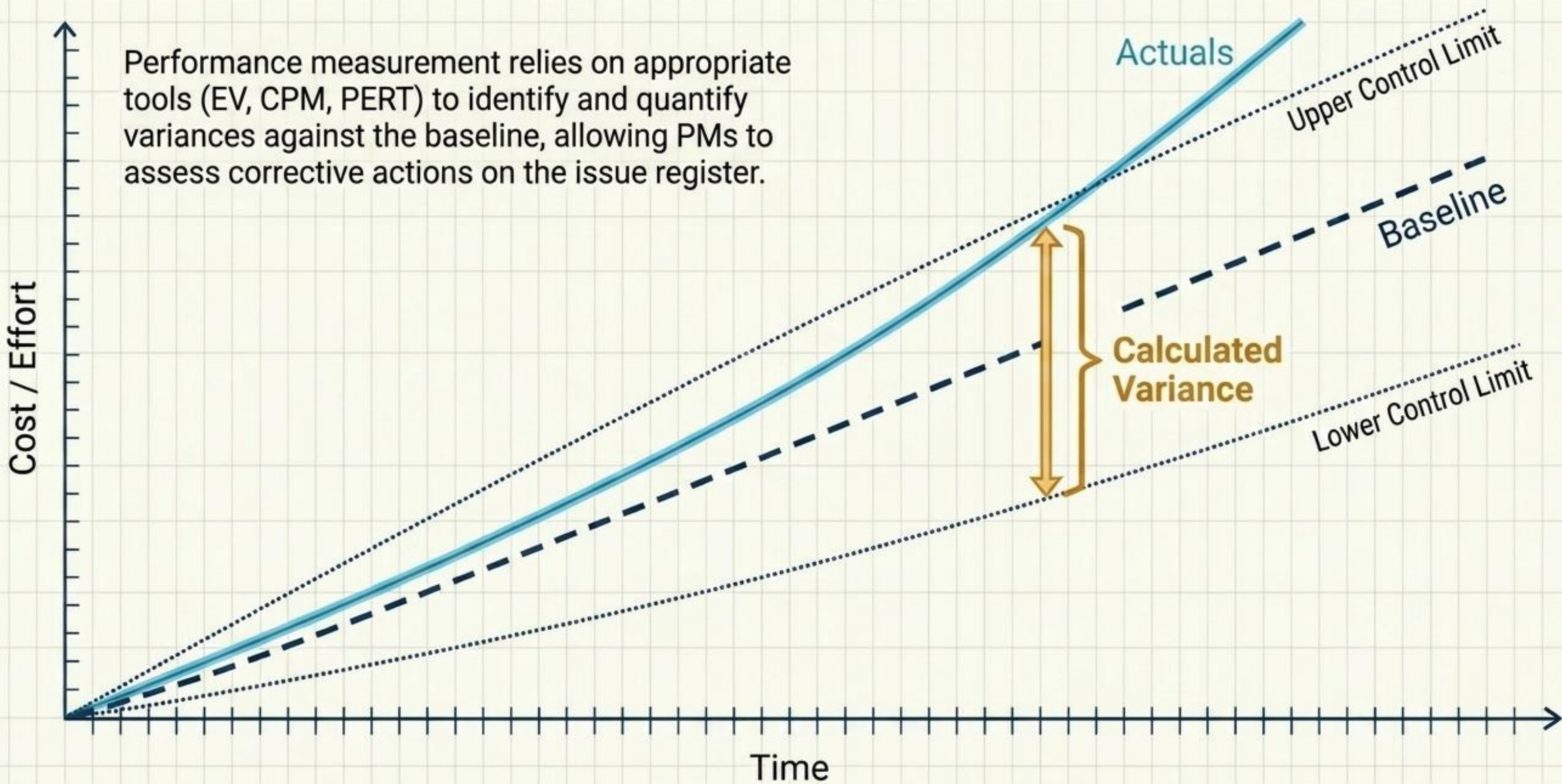
3. Adjust the Plan

Repeat affected planning processes to update baselines and document the new reality.

The Integrated Oversight Ecosystem

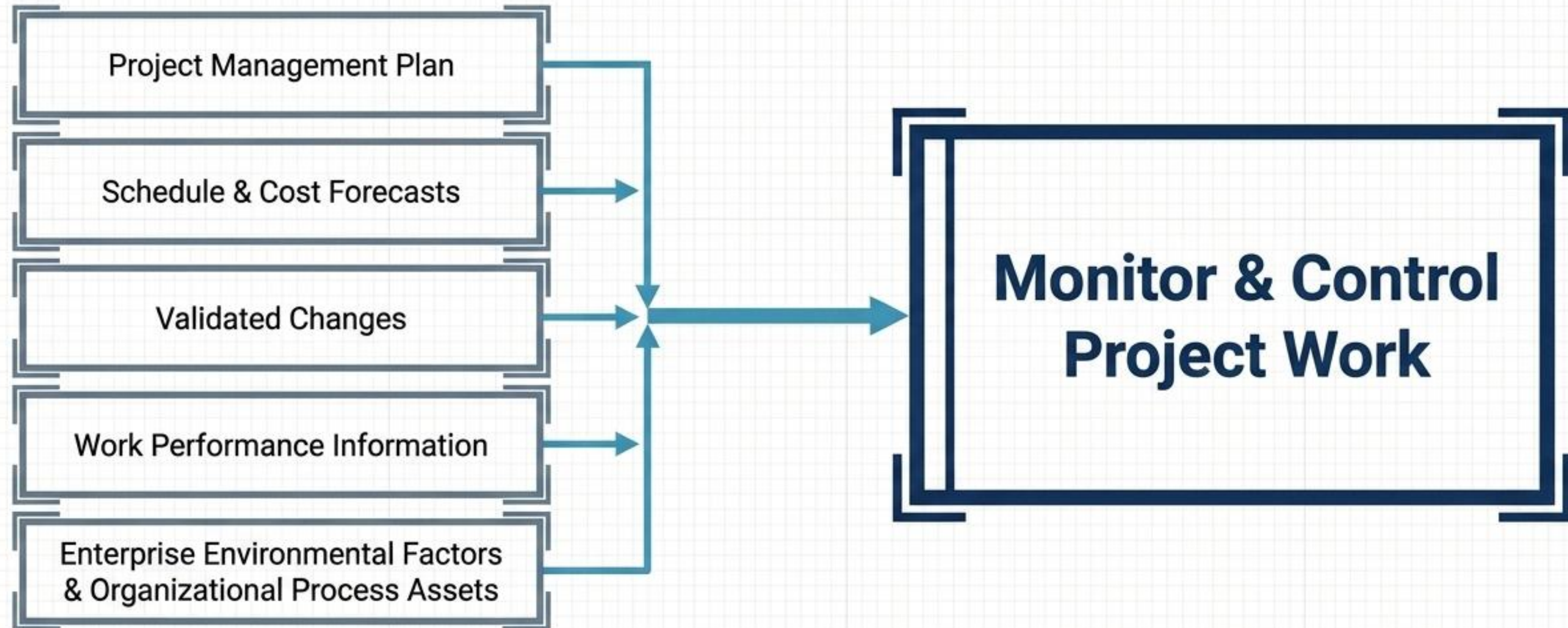


Quantifying the Deviation



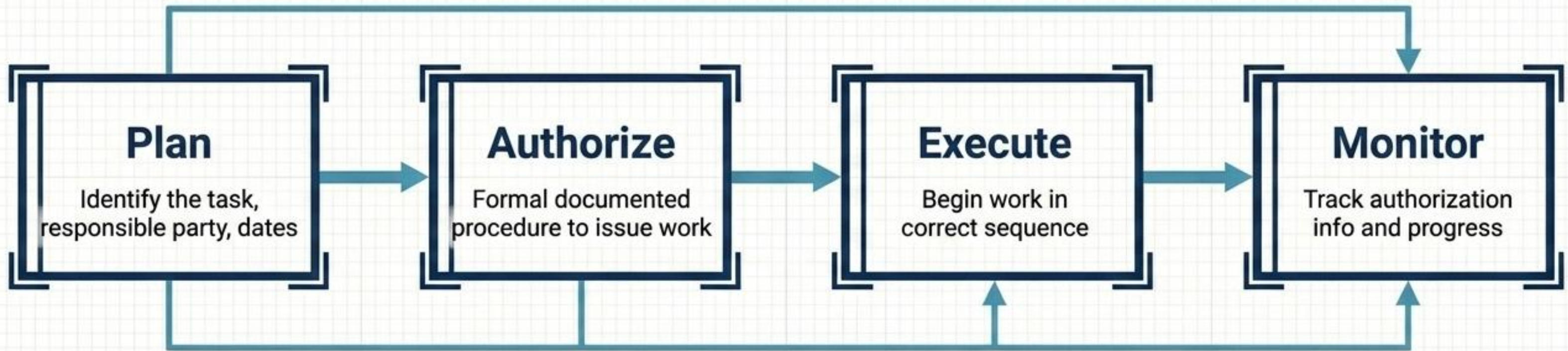
The System Inputs Architecture

Monitoring project work requires a constant stream of high-fidelity data. These inputs fuel the analytical techniques and expert judgment required to generate actionable change requests.



EEF Deep Dive: The Work Authorization Flow

A subsystem of the project management system that formally dictates how to authorize work and begin it in the right sequence at the right time.



Scalability: Large projects require formal, documented approval systems, while verbal instructions may suffice for smaller initiatives to avoid administrative overkill.

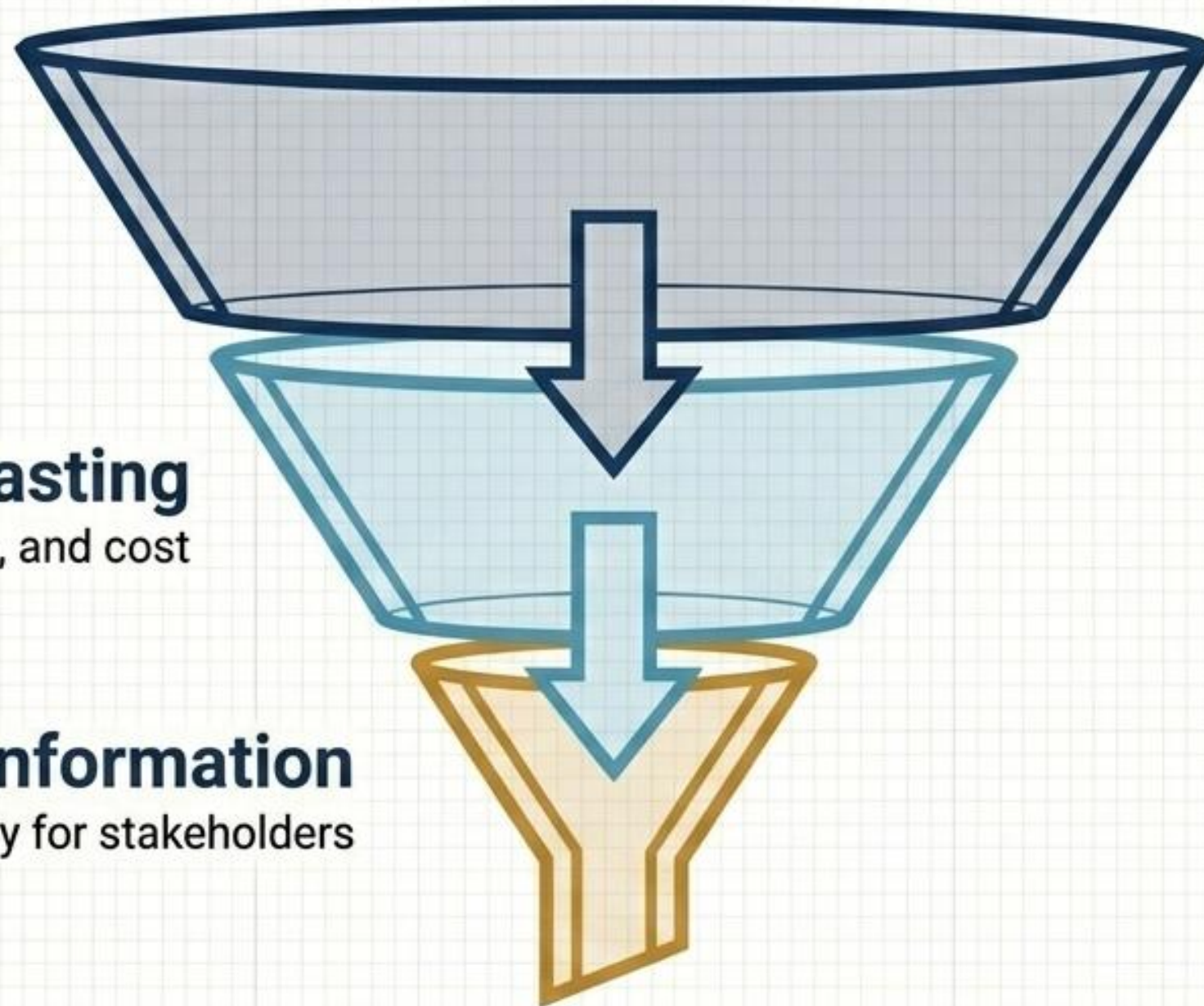
The Information Maturation Funnel

Data alone is insufficient. Work performance information is maintained during the Monitoring and Controlling processes to remedy wayward results and realign the future. This refined intelligence includes the status of deliverables, schedule/cost forecasts, and change request status.

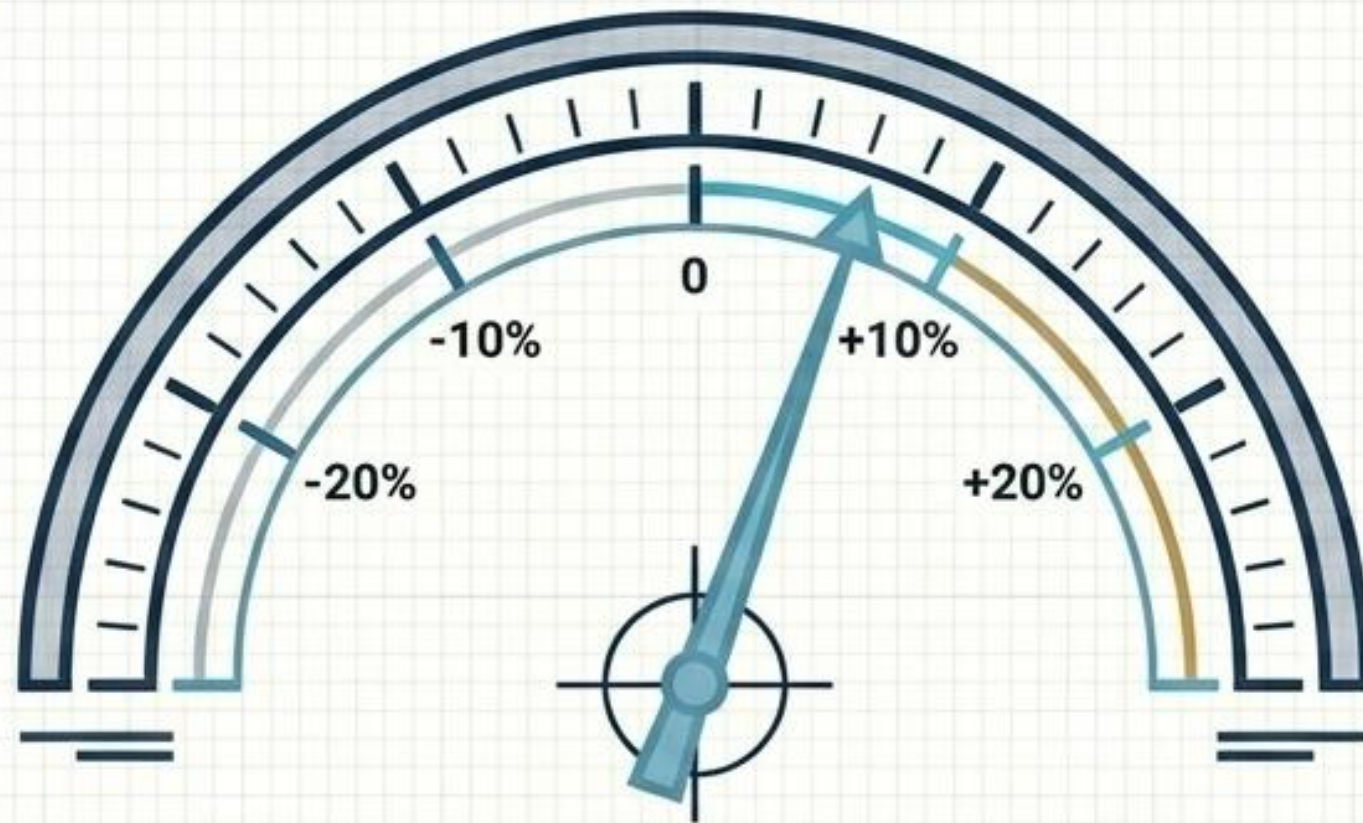
Performance Data
Raw work results from Executing processes

Analysis & Forecasting
Measuring against baselines, schedule, and cost

Work Performance Information
Actionable intelligence ready for stakeholders

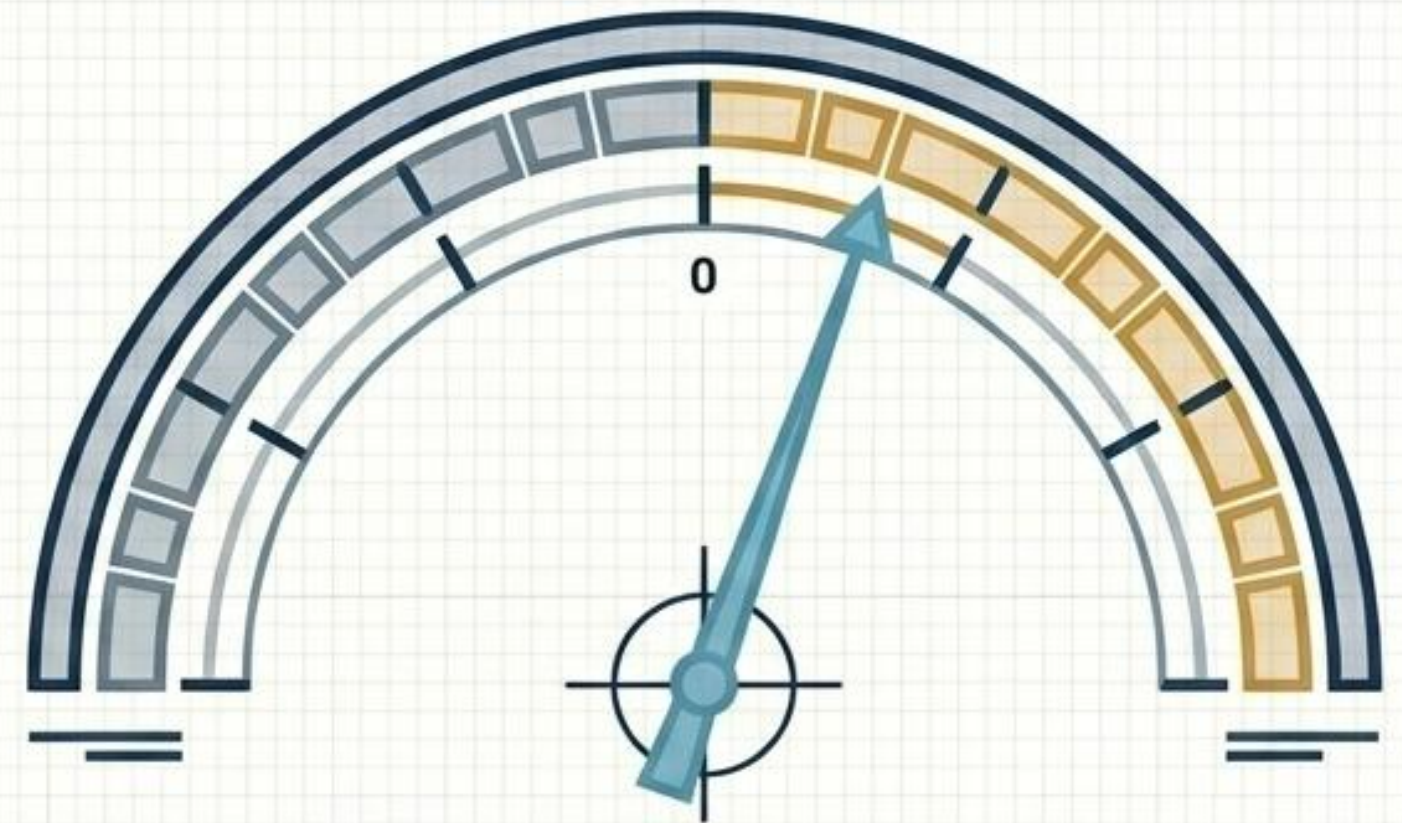


Predicting Project Trajectories



Schedule Forecasts

Compares schedule baseline against actual completion dates. Derived from the Control Schedule process to determine overall temporal progress.



Cost Forecasts

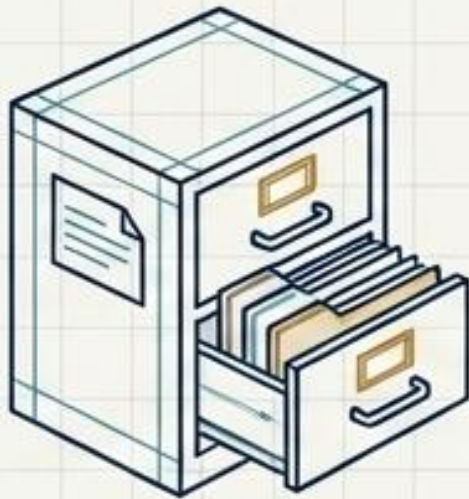
Compares anticipated spend against actual spend using Earned Value Management (EVM) formulas, such as Estimate to Complete (ETC) and Estimate at Completion (EAC). Derived from the Control Costs process.

The Forecasting Typology Matrix

			
Time Series Methods	Causal / Econometric	Judgmental Methods	Other Methods
Mechanism: Uses historical data to predict future performance.	Mechanism: Identifies variables that influence the forecast (e.g., interest rate drops).	Mechanism: Relies on opinions, intuition, and probability estimates.	Mechanism: Uses complex modeling and probabilistic forecasting.
Examples: Earned Value, Moving Average, Extrapolation, Trend Estimation.	Examples: Regression Analysis, Econometrics, ARMA.	Examples: Delphi Method, Composite Forecasts, Scenario Building.	Examples: Monte Carlo Simulation, Ensemble Forecasting.

The Diagnostic Toolkit

Organizational Process Assets (The Archive)



Historical rules and records that govern control, including financial controls, issue management procedures, process measurements, and lessons learned.

Tools & Techniques (The Application)



The mechanisms used to process data, including Expert Judgment, Analytical Techniques, Project Management Information Systems (PMIS), and structured Meetings.

The Continuous Control Loop

Project control is not a static event; it is a holistic, circular ecosystem. Raw data becomes insight, insight reveals variance, variance triggers changes, and approved changes recalibrate the baseline for the entire project.

